Computational Modeling - Assignment 1

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Github: https://github.com/Study-Group-2-0/Comp_modelling

The file is called “FINAL_Assignment1” and is available both as pdf and R-markdown.

Part 1

1. What's Riccardo's estimated knowledge of CogSci? What is the probability he knows more than chance (0.5)

(Anders) Figure 1.1 shows the posterior probability distribution of Riccardo’s estimated knowledge of CogSci given a uniform prior and evidence from Riccardo’s questionnaire. Grid approximation with a density of 20 was used. The red horizontal line shows the uniform prior.

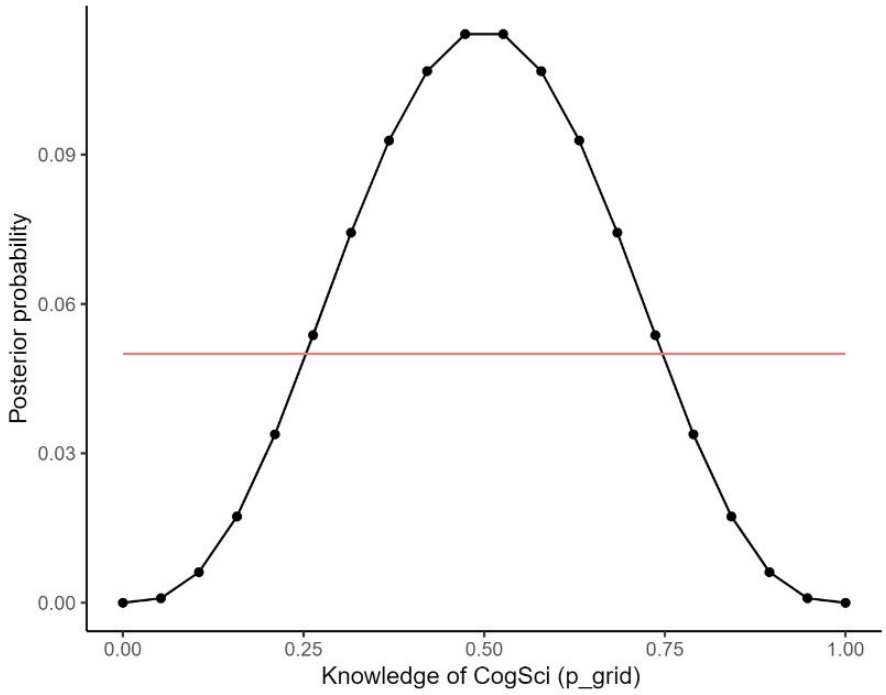

Figure 1.1 Riccardo’s estimated knowledge of CogSci using grid approximation. The red horizontal line shows the uniform prior.

Figure 1.2 shows the posterior probability distribution of Riccardo's knowledge of CogSci given a uniform prior and evidence from Riccardo's questionnaire. Riccardo's estimated knowledge of CogSci is not given by a single value but rather as two parameters describing the posterior probability distribution. The mean of the posterior probability distribution is 0.5 and the standard deviation is 0.2.

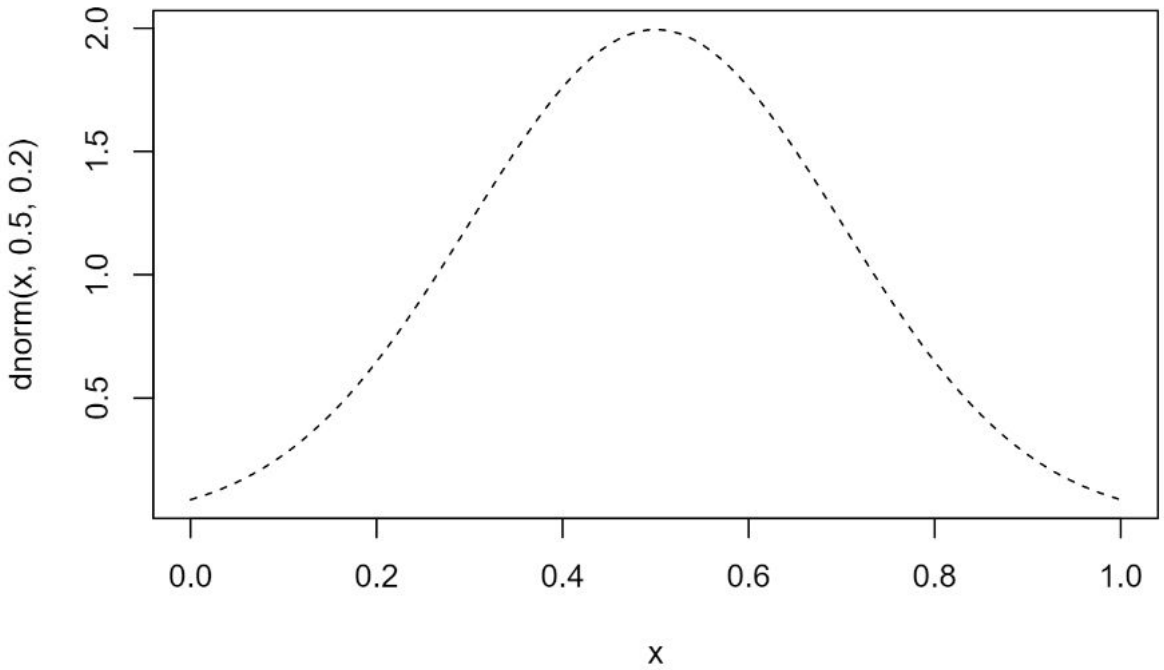

Figure 1.2 Riccardo's estimated knowledge of Cogsci using quadratic approximation.

The probability that Riccardo knows more than chance is 0.5 or 50 %.This also makes sense intuitively, because the distribution is centered with a mean of 0.5.

2. Estimate all the teachers' knowledge of CogSci. Who's best? Use grid approximation. Comment on the posteriors of Riccardo and Mikkel.

(Sigrid) In order to investigate the four teachers' posterior distribution, we used grid approximation. The grid consisted of 100 points between 0 and 1, and we used a uniform prior of 1. A tibble was created where the likelihood of each teacher were calculated and afterwards standardized in order to get the posterior probability for each of the 100 parameter values. The data was then transformed into a long format to plot it, and the final plot can be seen in figure 2, where each of the teachers posterior distributions are plotted together with the uniform prior (the dark blue horizontal line). The plot shows that the model on Kristian and Riccardo are not very certain about any parameter values. They both have a peak but they are very wide. This is most likely due to the small data provided for these two teachers. Riccardo answered 6 questions and Kristian 2. The model on Mikkel and Daina are, however, more certain about the parameter values. They both have a narrow pointy peak, that indicates that the model is relatively certain that the parameter value lies within the interval of the

peak. This is due to the bigger data provided to the model for these two teachers. Mikkel answered 132 questions and Daina 198 questions. The certainty's dependence on the magnitude of the data provided becomes clear when comparing Riccardo's and Mikkel's plots. They have both answered 50% of the questions correctly. However, Mikkel's posterior distribution is much more pointy and certain on the parameter values since Mikkel answered 132 questions and Riccardo only answered 6 questions.

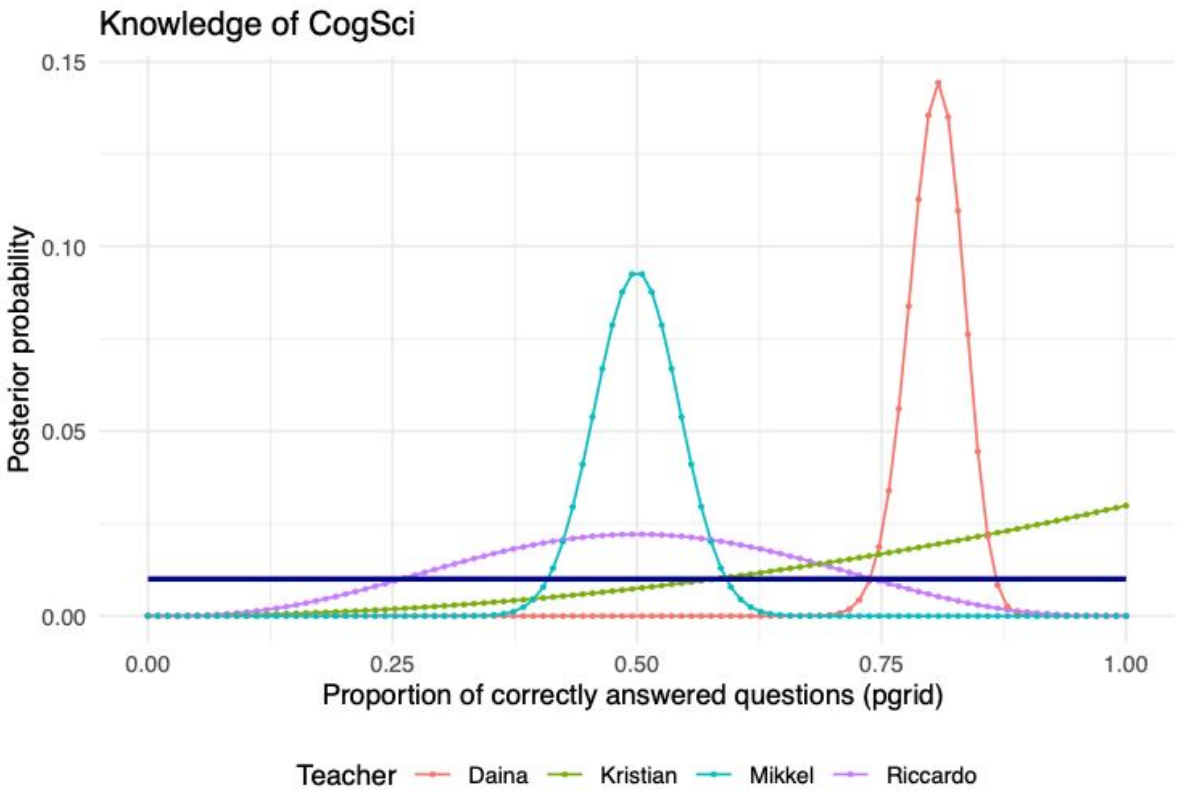

Figure 2. The posterior distributions of each of the teachers.

3. Change the prior. Given your teachers have all CogSci jobs, you should start with a higher appreciation of their knowledge: the prior is a normal distribution with a mean of 0.8 and a standard deviation of 0.2. Do the results change (and if so how)?

(Gustav) In figure 3, the prior (blue bold line) is lo longer flat compared to the posterior distributions made with a uniform prior (figure 2). The biggest difference can be found in the posterior distributions of Riccardo and Kristian, respectively. The standard deviation of the two distributions has become smaller. It could be that all of the distributions have become a bit more narrow (smaller

SD), but it is the most visible in the distributions with the least data, that is, the distributions of Riccardo and Kristian. Additionally, all of the distributions have moved closer to 1, which makes sense since the chosen prior has a mean of 0.8.

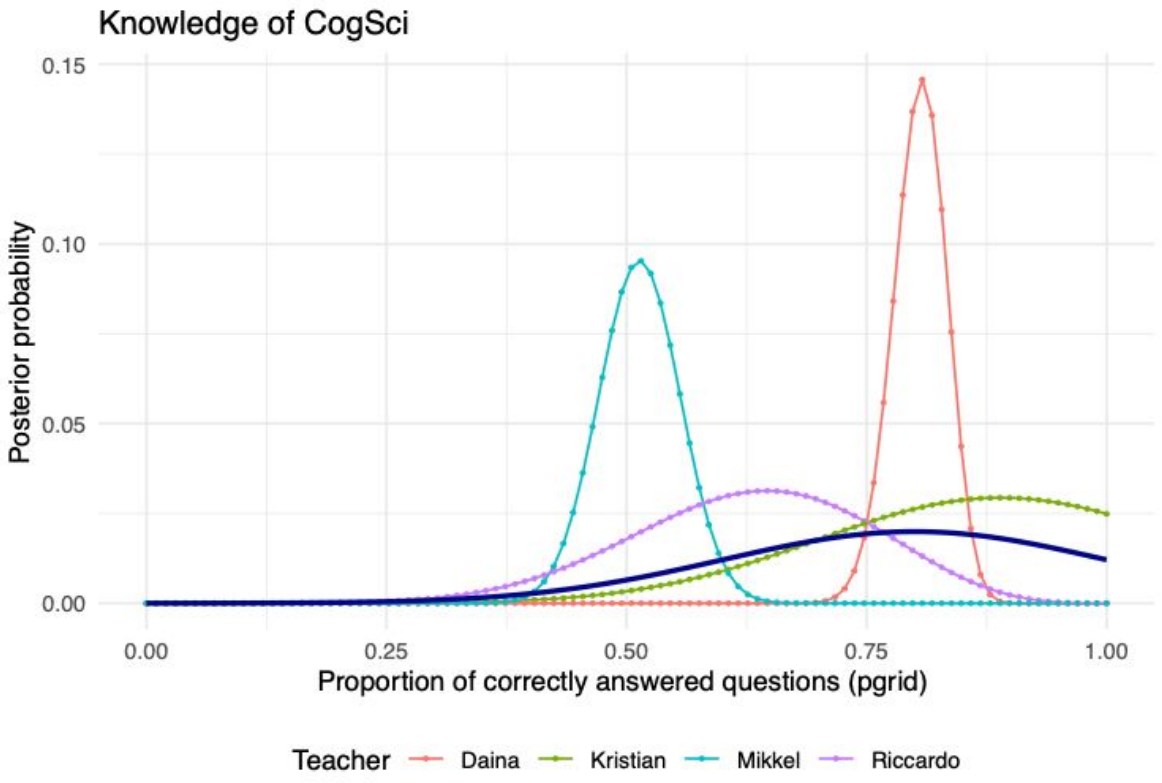

Figure 3: The posterior distributions of Daina, Kristian, Mikkel and Riccardo. The blue bold line is the prior.

4. You go back to your teachers and collect more data (multiply the previous numbers by 100). Calculate their knowledge with both a uniform prior and a normal prior with a mean of 0.8 and a standard deviation of 0.2. Do you still see a difference between the results? Why?

(Anders) As the amount of data increases, the distributions become more narrow. This means that given the data, a specific (and smaller) number of correct answers is more plausible. The prior does not have a great effect on the different distributions. This might be because we have a simple model and a lot of data (relatively). The prior will influence the distributions more, when there is a predictor variable and/or less data.

The posterior distributions with a normal prior can be found in figure 4.1 and the posterior distributions with a uniform prior can be found in figure 4.2.

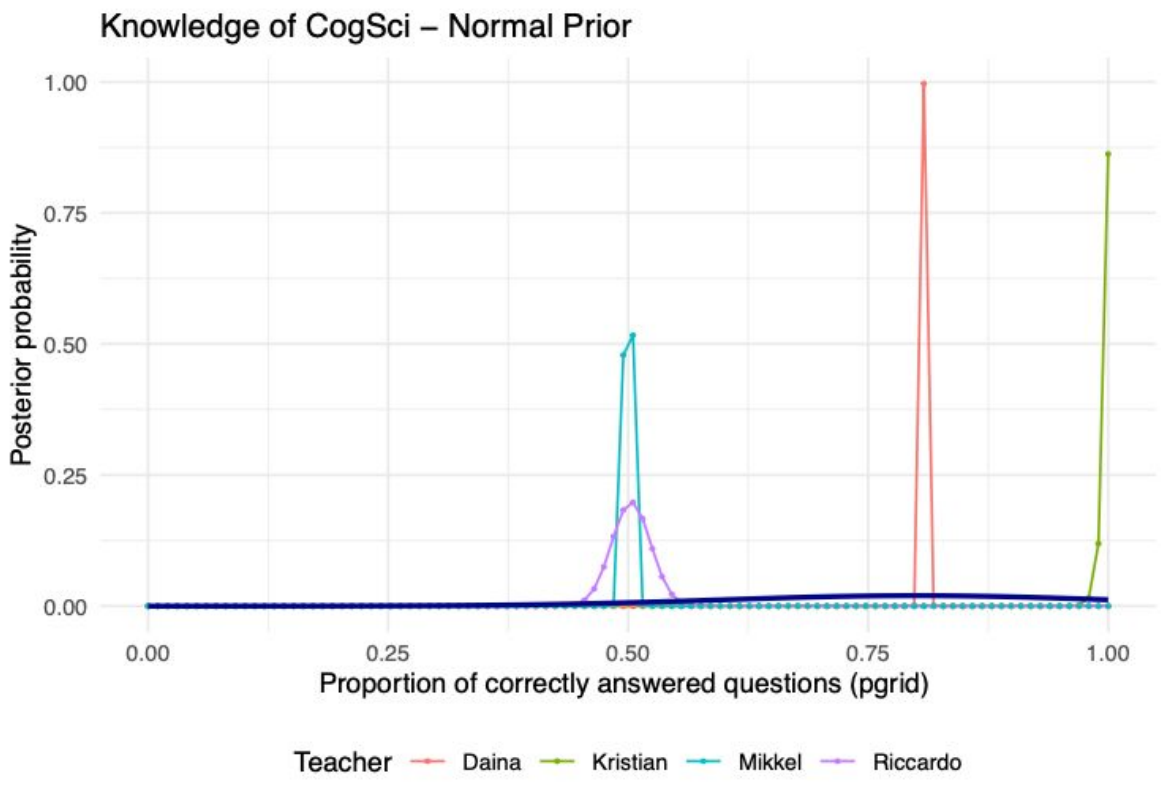

Figure 4.1: The posterior distributions made with a normal prior (mean of 0.8, standard deviation of 0.2).

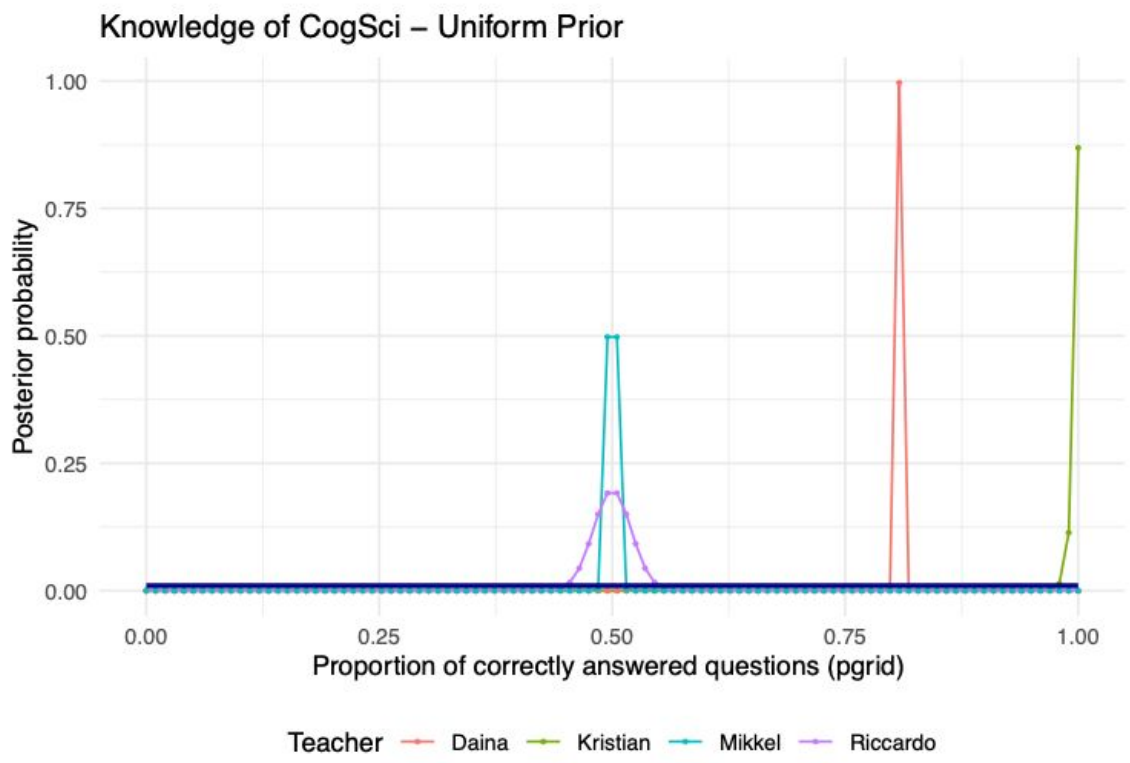

Figure 4.2: The posterior distributions made with a uniform prior.

5. Imagine you're a skeptic and think your teachers do not know anything about CogSci, given the content of their classes. How would you operationalize that belief?

(Gustav) We designed our skeptic prior by using a normal distribution with a mean of 0.5 and a SD of 0.2 on all grid values above 0.5. All grid values below 0.5 where given a prior value equal to the peak of the normal distribution. This prior predicted that all values below chance (the parameter value 0.5) were very plausible and as the parameter value exceeded chance level it gradually decreased in plausibility. The new posterior distribution of the teachers and the prior can be seen in figure 5. The skeptic prior had only influence on the posterior distributions of Kristian and Daina. They were both pushed towards the lower parameter values. However, the prior had no other obvious influence on the posterior distributions.

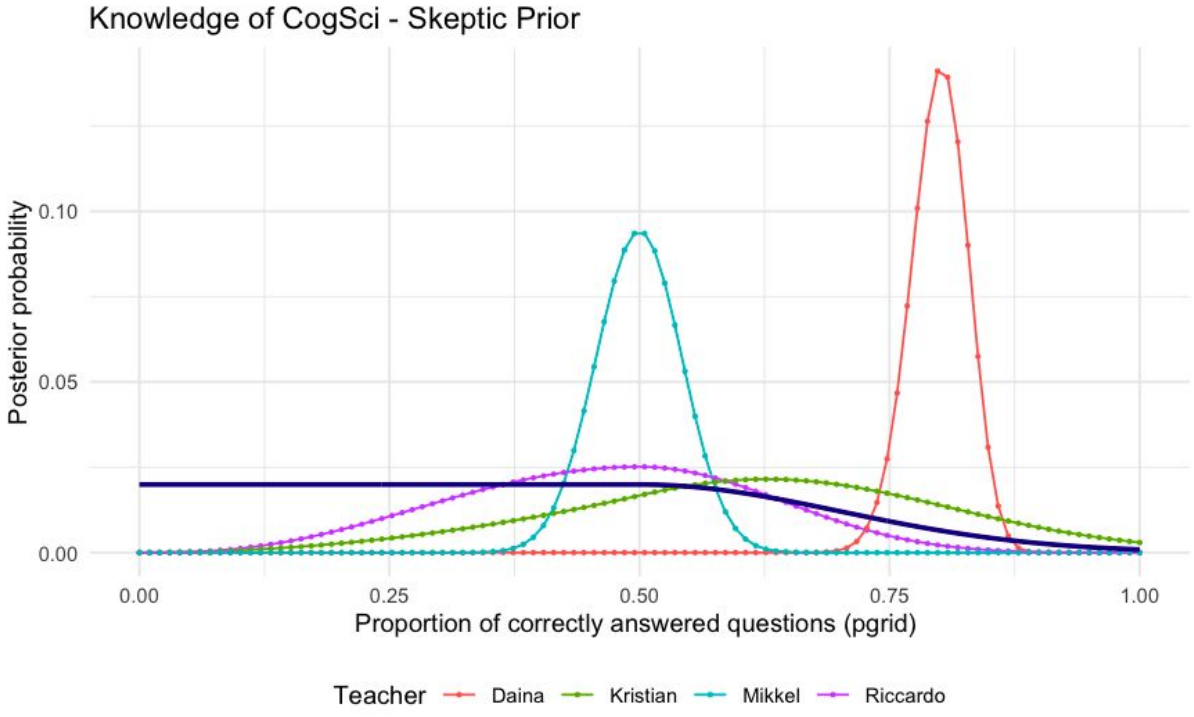

Figure 5. The posterior distributions of the teachers' knowledge of CogSci with a skeptic prior.

6. Optional question: Can you estimate the difference between Riccardo's estimated knowledge and that of each of the other teachers? Would you deem it credible (that is, would you believe that it is actually different)?

(Anders) To calculate the difference between the knowledge of Riccardo and the other teachers we sampled from the posterior probability distributions of the four teachers respectively, using a sampling rate of 1000. The result can be seen in figure 6.1. The sample distribution of Riccardo overlaps with all the other teachers to some extent. The question is whether this overlap is large enough to make Riccardo's distribution different from the others.

We ran a linear model to test whether Riccardo was indeed different from the other teachers using the following syntax:

Samples_Values ~ Teacher

Taken at face value, the result showed that Riccardo's knowledge of Cogsci was significantly different from all the other teachers ($p<0.001$ for all teachers, Beta_Daina = 0.62, Beta_Kristian = 0.17, Beta_Mikkel = -0.11). However we should not place confidence in these results since the data from Riccardo and Kristian is minimal (only 6 and 2 data points respectively!). To estimate whether the knowledge of the teachers are actually different we would therefore need to collect more data.

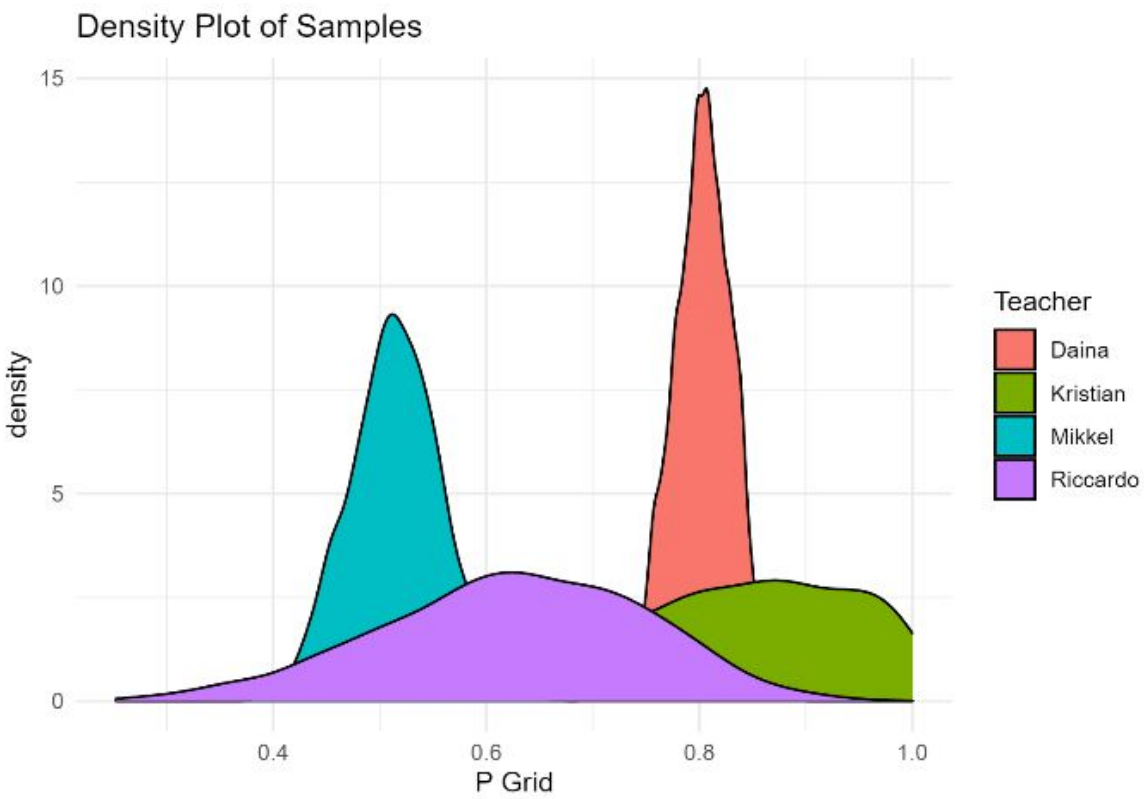

Figure 6.1: Density plot of samples drawn from the posterior probability distributions of the 4 teachers respectively.

Part 2

1. Write a paragraph discussing how assessment of prediction performance is different in Bayesian vs. frequentist models

(Sigrid)

In frequentist models, the prediction performance of a linear model can be calculated using RMSE (Root Mean Square Error). RMSE gives one single value as output, describing the distance from the predicted values to the actual values. It is thus a measure of prediction error or prediction performance.

In Bayesian models, the prediction performance of a model is not given by a single value, but rather by comparing posterior distributions with either the observed data or sampled data.

In general the prediction performance is not assessed by a single value but rather by a distribution.

In Bayesian models, there are at least two ways of assessing prediction performance. We will go through the two ways in the following sections.

Way 1:

One way is to use the posterior distribution with the old data as a prior for a new posterior distribution. The posterior distributions of the old and the new data, respectively, can then be compared visually (figure 7.1).

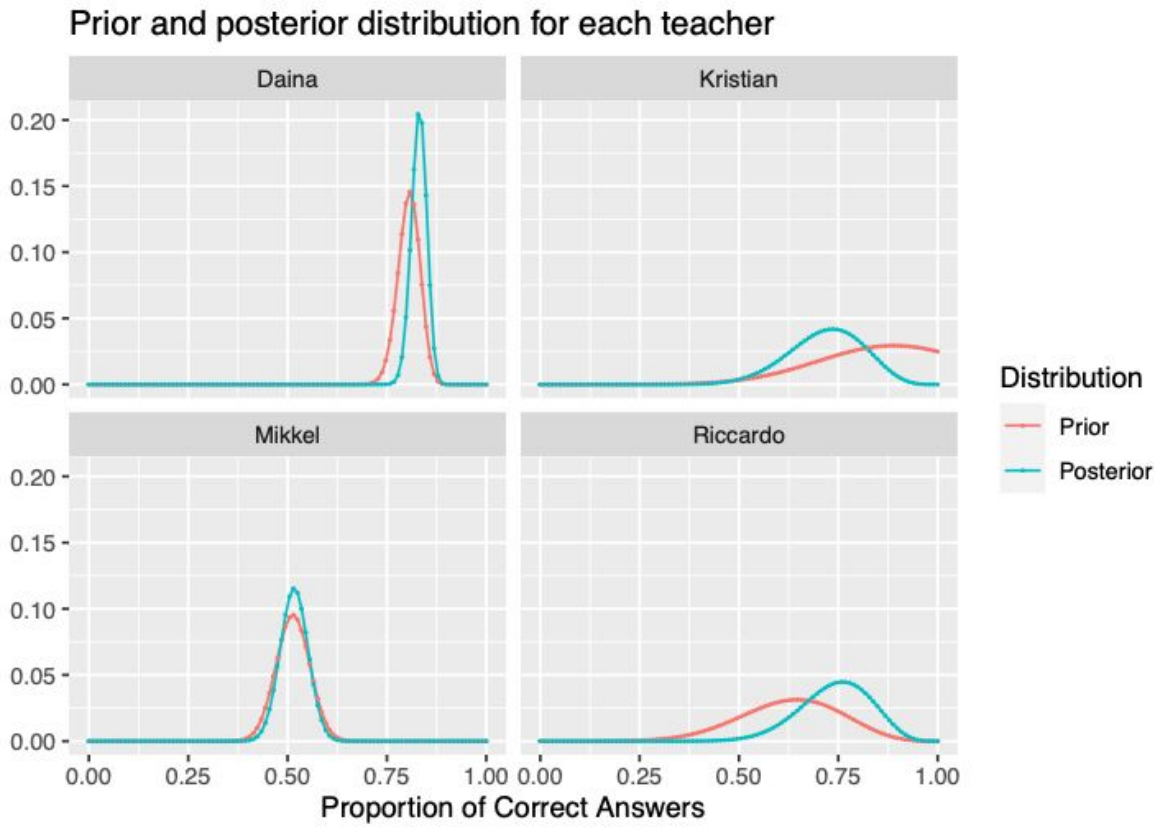

Figure 7.1. The posterior distributions of the old and the new data, respectively. 'Prior' is the post. dist. of the old data. 'Posterior' is the post. dist. of the new data, with the old post. dist. as a prior.

In order to investigate the distribution parameters, we sampled the posterior distributions of the old and the new data, respectively (sample size = 1000). Then, we compared the highest posterior density interval (HPDI) values of the different distributions (with the default interval if 89%). The change in HPDI-values across distributions for each teacher can be found in figure 7.2. The HPDI consists of two values for each distribution, a lower and an upper limit of the interval in which 89% of the values of the distribution can be found. It is seen that the posterior distributions of Riccardo and Kristian change the most, where Daina's and Mikkel's distribution only show a slight change.

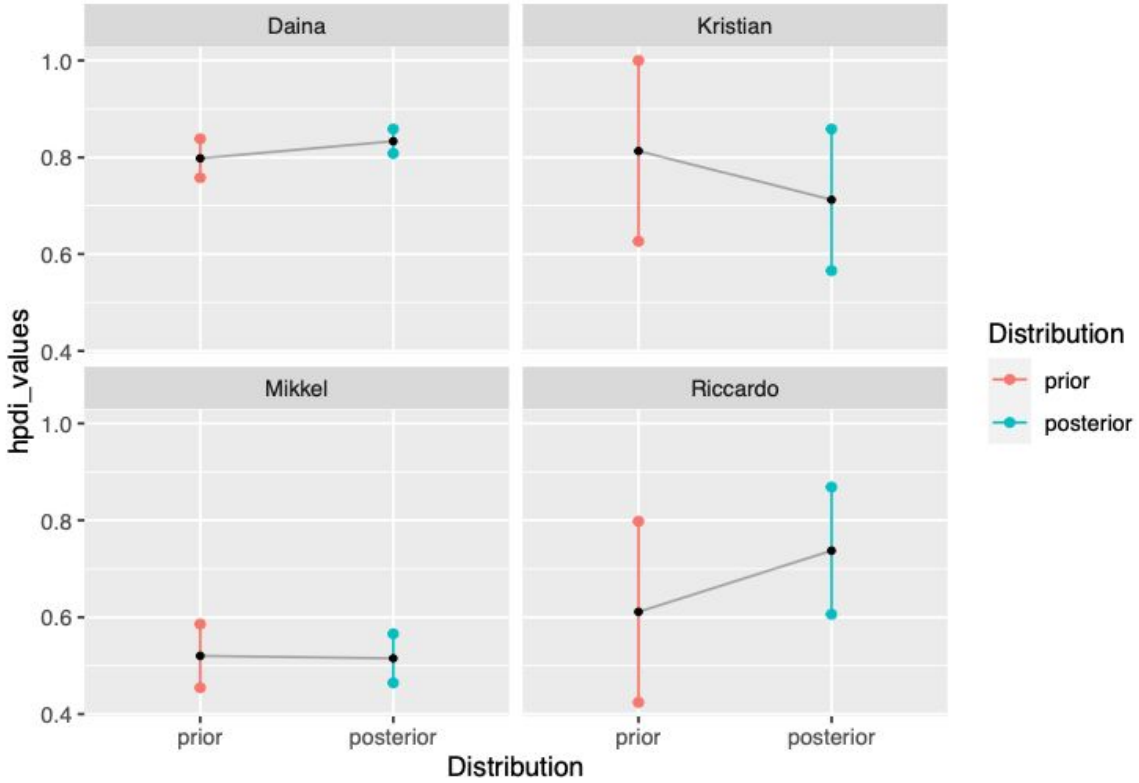

Figure 7.2: The HPDI-values of the different posterior distributions. 'Prior' is the post. dist. of the old data. 'Posterior' is the post. dist. of the new data with the old post. dist. as a prior.

In order to control our choice of prior, we compared the new posterior distribution (with the old post. dist. as prior) and a posterior distribution containing both the old and the new data. As expected, the distributions are identical (figure 7.3).

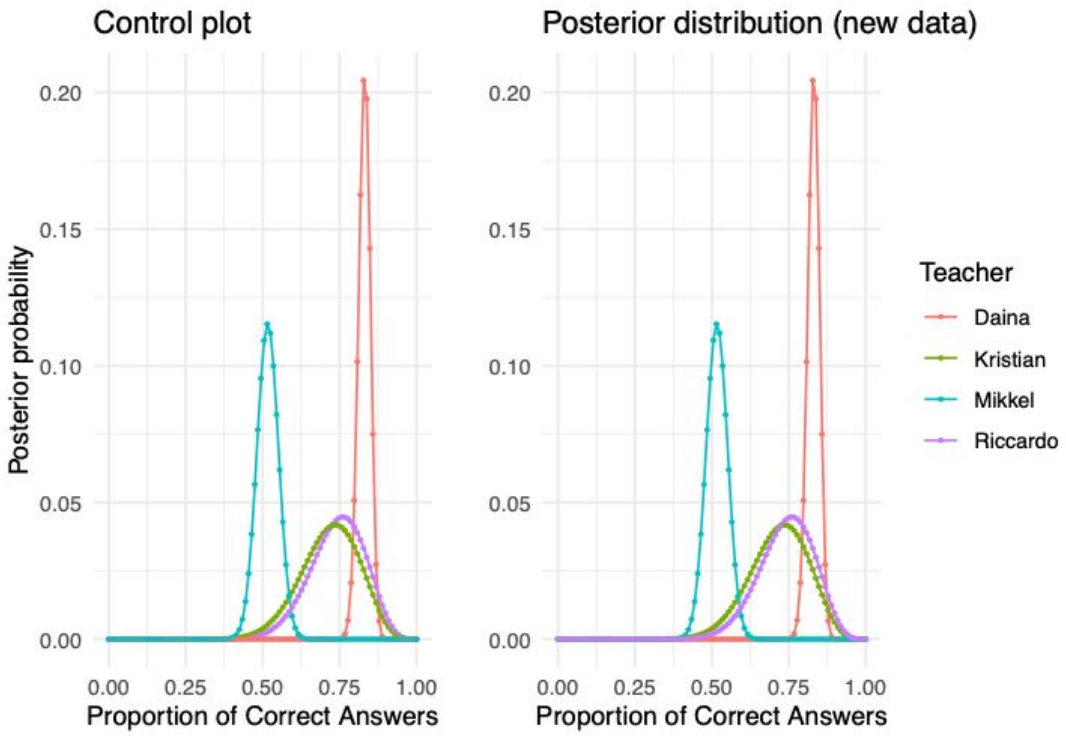

Figure 7.3. Left: Control plot of the posterior distributions of both old and new data with a normal prior. Right: The posterior distribution of the new data with the old post. dist. as the prior.

Way 2:

(Gustav) The second way to access prediction performance in a bayesian framework is to use the old data to generate a predictive posterior distribution that simulates the new data. This was done by first sampling from the old data and then using these samples to simulate how many correct answers the old data would predict if the teacher had been asked n questions. As n we used the number of questions the teacher had answered in the new data. The simulated results can be seen in figure 7.4. The highlighted column is the amount of correct answers the teacher actually got in the new data and the two blue lines indicates the parameter values for the highest posterior density interval (HPDI) with a probability of 50 %. This is the shortest interval of x-values within which 50 % of the simulations fell within. We choose an interval of 50 % hence it is more conservative than the default 89 %. The 50 % narrows the prediction field within which we accept correct predictions.

Figure 7.4 shows the predictive posterior of the four teachers respectively. In the new test, Riccardo had 9/10 correct answers. As seen in figure 7.4, this new datapoint lies outside the 50% interval of predictions. Furthermore the new scores of Kristian and Daina also lie outside the 50% interval of predictions. The only teacher whose new score lies within the 50% interval is Mikkel(Ræv). In general, the predictions of the models are not very precise at capturing the new test-scores. However,

the highest density intervals provided by the HDPI function from the rethinking package seems odd. In the plot of the Kristian's distribution it would theoretically make sense that the HDPI would be in the far right of the plot hence this is where the most observations are. However, the HDPI interval is not in the far right indicating that there is some mistake. We have not been able to detect this error but hope you can help enlighten us :).

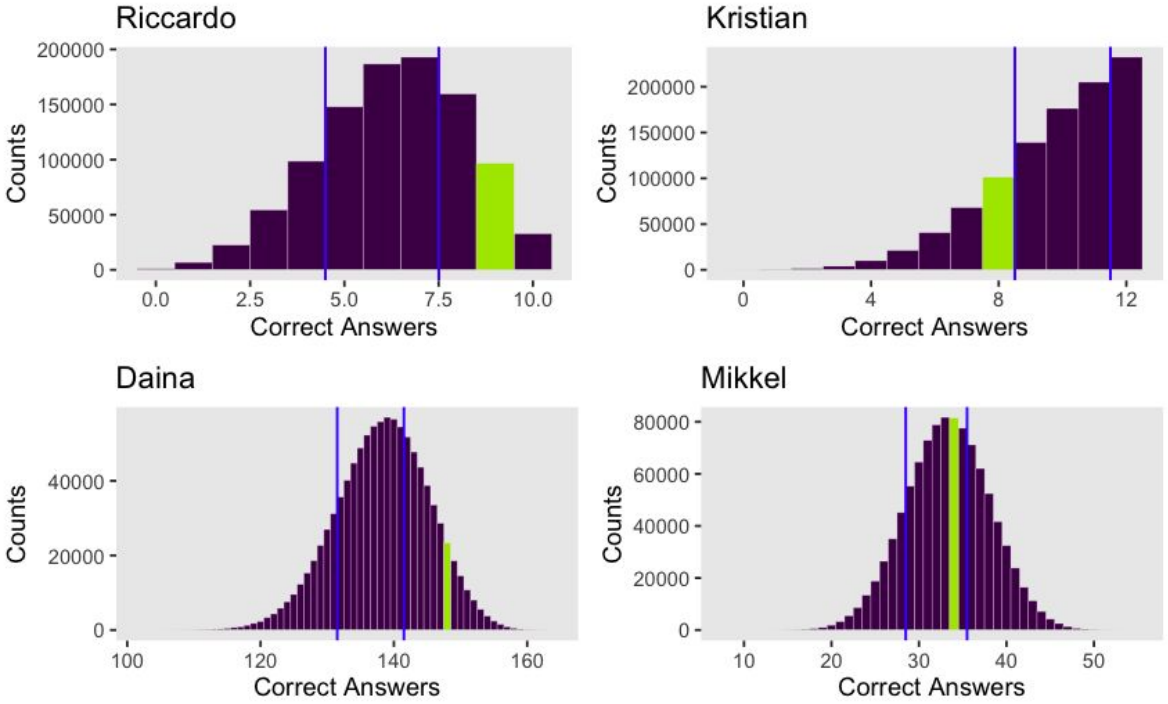

Figure 7.4. The predictive posterior distributions of the teachers' with sample size = 1.000000.